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TREND-BASED MOCK PAPER

GATE Computer Science and Information Technology

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Computer Science and Information Technology
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [2-MCQ] [2 marks]

Antonym of 'ephemeral':

- (A) Transient
- (B) Permanent
- (C) Brief
- (D) Short

Section B: Computer Science and Information Technology

Q2. [2-NAT] [2 marks]

BFS uses:

(Numerical Answer Type — Enter the numerical value)

Q3. [2-MCQ] [2 marks]

Merge sort space:

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n^2)$

Q4. [2-NAT] [2 marks]

Worst-case time complexity of merge sort:

(Numerical Answer Type — Enter the numerical value)

Q5. [2-MCQ] [2 marks]

LCS uses:

- (A) Greedy

- (B) Divide and Conquer
- (C) Dynamic Programming
- (D) Backtracking

Q6. [3-MCQ] [4 marks]

The time complexity of Dijkstra's algorithm using a binary heap is:

- (A) $O(V^2)$
- (B) $O(E \log V)$
- (C) $O(V \log V + E)$
- (D) $O(V \log V)$

Q7. [2-MCQ] [2 marks]

Kruskal uses:

- (A) DFS
- (B) BFS
- (C) Union-Find
- (D) Dijkstra

Q8. [2-NAT] [2 marks]

Time complexity of binary search:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-MCQ] [1 mark]

A minimum vertex cover in a bipartite graph equals:

- (A) Maximum matching
- (B) Size of maximum matching
- (C) Minimum edge cover
- (D) Maximum independent set

Q10. [1-NAT] [1 mark]

Kruskal's algorithm uses for efficiency:

(Numerical Answer Type — Enter the numerical value)

Q11. [1-MCQ] [1 mark]

BFS tree from source s: tree edges form:

- (A) Shortest path tree
- (B) Minimum spanning tree
- (C) DFS tree
- (D) Arborescence

Q12. [1-MCQ] [1 mark]

Bellman-Ford can detect:

- (A) Positive cycles
- (B) Negative cycles
- (C) Zero cycles
- (D) All cycles

Q13. [1-MCQ] [1 mark]

Min number of comparisons to find both min and max of n elements:

- (A) $2n-2$
- (B) $\text{ceil}(3n/2)-2$

- (C) $n-1$
- (D) $2n-3$

Q14. [1-NAT] [1 mark]

Amortized cost of splay tree operation:

(Numerical Answer Type — Enter the numerical value)

Q15. [1-MCQ] [1 mark]

Topological sort is possible only for:

- (A) Cyclic graphs
- (B) Acyclic graphs
- (C) Complete graphs
- (D) Bipartite graphs

Q16. [1-MCQ] [1 mark]

Dijkstra fails with:

- (A) Positive edges
- (B) Zero edges
- (C) Negative edges
- (D) No cycles

Q17. [2-MSQ] [2 marks]

TLB stands for:

- (A) Translation Lookaside Buffer
- (B) Time Local Buffer
- (C) Total Logic Block
- (D) None

Q18. [3-MCQ] [4 marks]

Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2,3,4?

- (A) FIFO
- (B) LRU
- (C) Optimal
- (D) None of these

Q19. [2-NAT] [2 marks]

Round-robin uses:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-NAT] [2 marks]

Which of the following is NOT a necessary condition for deadlock?

(Numerical Answer Type — Enter the numerical value)

Q21. [2-NAT] [2 marks]

Inode contains:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

Semaphore with value 1 is:

- (A) Counting
- (B) Binary (mutex)
- (C) Monitor
- (D) Lock variable

Q23. [2-MCQ] [2 marks]

Banker's algorithm is for:

- (A) Detection
- (B) Avoidance
- (C) Prevention
- (D) Recovery

Q24. [1-MCQ] [1 mark]

SCAN disk scheduling is also called:

- (A) FCFS
- (B) Elevator algorithm
- (C) SSTF
- (D) C-SCAN

Q25. [2-MCQ] [2 marks]

Probability of at least one head in 3 coin tosses:

- (A) $1/8$
- (B) $3/8$
- (C) $7/8$
- (D) $1/2$

Q26. [2-NAT] [2 marks]

Derivative of x^3 at $x=2$:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-NAT] [2 marks]

Determinant of 2×2 identity:

(Numerical Answer Type — Enter the numerical value)

Q28. [2-NAT] [2 marks]

Mean of binomial $n=10, p=0.5$:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-NAT] [2 marks]

Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

(Numerical Answer Type — Enter the numerical value)

Q30. [1-MCQ] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q31. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q32. [1-NAT] [1 mark]

The Laplace transform of $t \cdot e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q33. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q34. [1-NAT] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q35. [1-NAT] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-MCQ] [2 marks]

OSPF is:

- (A) Distance vector
- (B) Link state
- (C) Path vector
- (D) Hybrid

Q37. [2-MCQ] [2 marks]

Asymmetric encryption:

- (A) AES
- (B) DES
- (C) RSA
- (D) 3DES

Q38. [2-MCQ] [2 marks]

Which protocol operates at the application layer of the TCP/IP model?

- (A) TCP
- (B) UDP
- (C) HTTP
- (D) IP

Q39. [2-MCQ] [2 marks]

CSMA/CD is used in:

- (A) Token Ring

- (B) Ethernet
- (C) WiFi
- (D) Bluetooth

Q40. [2-MCQ] [2 marks]

TCP connection establishment uses:

- (A) 2-way
- (B) 3-way handshake
- (C) 4-way
- (D) 1-way

Q41. [2-MCQ] [2 marks]

Protocol to find MAC from IP:

- (A) DNS
- (B) ARP
- (C) DHCP
- (D) ICMP

Q42. [1-NAT] [1 mark]

Class C IP 200.10.20.0 with /28 mask. Max subnets:

(Numerical Answer Type — Enter the numerical value)

Q43. [1-MCQ] [1 mark]

TCP ssthresh initially:

- (A) 0
- (B) 64 KB
- (C) 65536
- (D) Infinity

Q44. [1-MCQ] [1 mark]

In Go-Back-N, sender window size = 7. Sequence numbers needed:

- (A) 7
- (B) 8
- (C) 14
- (D) 6

Q45. [2-NAT] [2 marks]

Avg hash lookup:

(Numerical Answer Type — Enter the numerical value)

Q46. [2-MCQ] [2 marks]

Infix to postfix uses:

- (A) Queue
- (B) Stack
- (C) Tree
- (D) Graph

Q47. [2-MCQ] [2 marks]

Inorder traversal of BST gives:

- (A) Sorted order
- (B) Reverse sorted

- (C) Level order
- (D) Random

Q48. [2-MCQ] [2 marks]

What is the time complexity of inserting at the beginning of a linked list?

- (A) $O(n)$
- (B) $O(1)$
- (C) $O(\log n)$
- (D) $O(n^2)$

Q49. [1-MCQ] [1 mark]

3-to-8 decoder has how many output lines?

- (A) 3
- (B) 8
- (C) 11
- (D) 6

Q50. [1-NAT] [1 mark]

2K x 8 ROM has how many address lines?

(Numerical Answer Type — Enter the numerical value)

Q51. [1-MCQ] [1 mark]

Interrupt-driven I/O vs DMA:

- (A) DMA is faster for single byte
- (B) DMA is better for block transfer
- (C) Interrupt is better for large data
- (D) Both same

Q52. [1-MCQ] [1 mark]

Cache 64 blocks, 4-way set assoc, block 16B. Address 200 maps to:

- (A) Set 12, offset 8
- (B) Set 8, offset 12
- (C) Set 4, offset 8
- (D) Set 12, offset 0

Q53. [1-MCQ] [1 mark]

4-bit Johnson counter sequences through:

- (A) 4
- (B) 8
- (C) 10
- (D) 16

Q54. [1-NAT] [1 mark]

Cache 64 blocks, 4-way set assoc, block 16B. Address 200 maps to:

(Numerical Answer Type — Enter the numerical value)

Q55. [2-MCQ] [2 marks]

Flip-flop stores:

- (A) 1 bit
- (B) 2 bits
- (C) 4 bits

(D) 8 bits

Q56. [1-NAT] [1 mark]

Full adder has how many inputs:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-MCQ] [1 mark]

The number of Boolean functions of n variables is:

- (A) 2^n
- (B) $2^{(2^n)}$
- (C) $2^{(n^2)}$
- (D) n^2

Q58. [1-MCQ] [1 mark]

Which language is regular?

- (A) Equal 0s and 1s
- (B) Strings ending in ab
- (C) Palindromes
- (D) $a^n b^n$

Q59. [1-NAT] [1 mark]

Halting problem:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

PDA recognizes:

- (A) Regular
- (B) Context-free
- (C) Context-sensitive
- (D) Recursive

Q61. [1-NAT] [1 mark]

Transaction atomicity means:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

B+ tree leaf nodes are linked for:

- (A) Faster insert
- (B) Range queries
- (C) Faster delete
- (D) Balance

Q63. [1-NAT] [1 mark]

R(A,B,C,D) with FDs: AB->C, C->D, D->A. Candidate keys:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

3-address code operands:

- (A) 1
- (B) 2

- (C) 3
- (D) 4

Q65. [1-NAT] [1 mark]

Lexical analyzer outputs:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	2MCQ	B	2
2	2MCQ	C	2
3	2NAT	B	2
4	2NAT	B	2
5	2NAT	B	2
6	2NAT	C	2
7	2NAT	C	2
8	2MCQ	B	2
9	2MCQ	A	2
10	2MCQ	B	2
11	2NAT	B	2
12	2MCQ	C	2
13	2NAT	B	2

14	2MCQ	C	2
15	3MCQ	C	4
16	2MCQ	C	2
17	2NAT	B	2
18	2MSQ	A	2
19	3MCQ	C	4
20	2NAT	A	2
21	2NAT	C	2
22	2NAT	B	2
23	2MCQ	B	2
24	2MCQ	B	2
25	2MCQ	B	2
26	2MCQ	C	2
27	2MCQ	C	2
28	2MCQ	B	2

29	2MCQ	B	2
30	2MCQ	B	2
31	2MCQ	A	2
32	1NAT	B	1
33	1MCQ	B	1
34	1MCQ	B	1
35	1NAT	B	1
36	1MCQ	B	1
37	1MCQ	C	1
38	1NAT	A	1
39	1MCQ	A	1
40	1MCQ	B	1
41	1MCQ	B	1
42	1MCQ	A	1
43	1NAT	B	1

44	1NAT	B	1
45	1MCQ	A	1
46	1NAT	C	1
47	1NAT	B	1
48	1MCQ	B	1
49	1MCQ	C	1
50	1MCQ	B	1
51	1MCQ	B	1
52	1NAT	B	1
53	1MCQ	B	1
54	1MCQ	A	1
55	1MCQ	B	1
56	1NAT	A	1
57	1MCQ	B	1
58	1MCQ	B	1

59	1MCQ	B	1
60	1NAT	A	1
61	1MCQ	C	1
62	1NAT	B	1
63	1MCQ	B	1
64	1NAT	B	1
65	1NAT	B	1

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q2.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Probability | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: moderate

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Probability | Difficulty: easy

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: difficult

Q7.

Answer: C

Solution:

Correct Answer: C

Subject: Programming and Data Structures | Topic: Hashing | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Programming and Data Structures | Topic: Stacks | Difficulty: difficult

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Programming and Data Structures | Topic: Trees | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Programming and Data Structures | Topic: Linked List | Difficulty: moderate

Q11.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Graph Algorithms | Difficulty: difficult

Q12.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Divide and Conquer | Difficulty: moderate

Q13.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Sorting | Difficulty: easy

Q14.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Dynamic Programming | Difficulty: moderate

Q15.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Graph Algorithms | Difficulty: easy

Q16.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Greedy | Difficulty: moderate

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Complexity Analysis | Difficulty: easy

Q18.

Answer: A

Solution:

Correct Answer: A

Subject: Operating Systems | Topic: Virtual Memory | Difficulty: easy

Q19.

Answer: C

Solution:

Correct Answer: C

Subject: Operating Systems | Topic: Memory Management | Difficulty: easy

Q20.

Answer: A

Solution:

Correct Answer: A

Subject: Operating Systems | Topic: Process Scheduling | Difficulty: moderate

Q21.

Answer: C

Solution:

Correct Answer: C

Subject: Operating Systems | Topic: Deadlocks | Difficulty: easy

Q22.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: File Systems | Difficulty: moderate

Q23.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: Synchronization | Difficulty: difficult

Q24.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: Deadlocks | Difficulty: moderate

Q25.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Routing | Difficulty: moderate

Q26.

Answer: C

Solution:

Correct Answer: C

Subject: Computer Networks | Topic: Security | Difficulty: moderate

Q27.

Answer: C

Solution:

Correct Answer: C

Subject: Computer Networks | Topic: Application Layer | Difficulty: moderate

Q28.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Data Link Layer | Difficulty: moderate

Q29.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Transport Layer | Difficulty: moderate

Q30.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Network Layer | Difficulty: difficult

Q31.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Logic | Topic: Sequential Circuits | Difficulty: moderate

Q32.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Logic | Topic: Combinational Circuits | Difficulty: easy

Q33.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Logic | Topic: Boolean Algebra | Difficulty: moderate

Q34.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Regular Languages | Difficulty: easy

Q35.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Decidability | Difficulty: difficult

Q36.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Context Free Languages | Difficulty: moderate

Q37.

Answer: C

Solution:

Correct Answer: C

Subject: Compiler Design | Topic: Code Generation | Difficulty: difficult

Q38.

Answer: A

Solution:

Correct Answer: A

Subject: Compiler Design | Topic: Lexical Analysis | Difficulty: moderate

Q39.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: B

Solution:

Koenig's theorem: min vertex cover size = max matching size in bipartite graphs.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q41.

Answer: B

Solution:

3 inputs decode to 8 outputs.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: difficult

Q42.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: B

Solution:

/28 means 4 bits for subnetting from /24. $2^4 = 16$ subnets. Each has 16 addresses, 14 usable hosts.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: moderate

Q44.

Answer: B

Solution:

Kruskal: sort edges $O(E \lg E)$ + DSU $O(E \cdot \alpha(V))$. DSU is essential for cycle detection.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q45.

Answer: A

Solution:

BFS: discovers vertices level by level. Tree edges give shortest paths (unweighted graph).

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q46.

Answer: C

Solution:

$2K = 2048$ locations. Need $\log_2 = 11$ address lines.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: moderate

Q47.

Answer: B

Solution:

Atomicity: transaction either completes fully or has no effect. No partial application.

Subject: DBMS | Topic: DBMS | Difficulty: easy

Q48.

Answer: B

Solution:

Bellman-Ford: after $V-1$ relaxations, one more detects negative cycles.

Subject: Algorithms | Topic: Algorithms | Difficulty: easy

Q49.

Answer: C

Solution:

Initial ssthresh = 65536 bytes (64 KB). cwnd starts at 1-4 MSS.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: moderate

Q50.

Answer: B

Solution:

B+ tree: all data at leaves, linked sequentially. Enables efficient range scans without root-to-leaf traversal.

Subject: DBMS | Topic: DBMS | Difficulty: easy

Q51.

Answer: B

Solution:

Process pairs: $n/2$ comps to compare each pair, then compare winners to max and losers to min. Total: $3n/2 - 2$.

Subject: Algorithms | Topic: Algorithms | Difficulty: difficult

Q52.

Answer: B

Solution:

Splay tree: $O(\log n)$ amortized per operation via access lemma.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q53.

Answer: B

Solution:

DMA: direct memory access. Better for block transfers (no CPU per byte). Interrupt per byte.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: moderate

Q54.

Answer: A

Solution:

Block addr = $200/16 = 12$. Sets = $64/4 = 16$. Set = $12 \bmod 16 = 12$. Offset = $200 \bmod 16 = 8$.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: moderate

Q55.

Answer: B

Solution:

SCAN: head moves in one direction servicing all requests, then reverses. Like elevator.

Subject: Operating Systems | Topic: Operating Systems | Difficulty: moderate

Q56.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: B

Solution:

Topological sort requires DAG (Directed Acyclic Graph).

Subject: Algorithms | Topic: Algorithms | Difficulty: difficult

Q58.

Answer: B

Solution:

Johnson (4-bit): 0000, 0001, 0011, 0111, 1111, 1110, 1100, 1000 = 8 states.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: moderate

Q59.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q60.

Answer: A

Solution:

Block addr = $200/16 = 12$. Sets = $64/4 = 16$. Set = $12 \bmod 16 = 12$. Offset = $200 \bmod 16 = 8$.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: easy

Q61.

Answer: C

Solution:

Dijkstra: greedy, extracts min. Fails with negative edges (may extract non-optimal path first). Use Bellman-Ford.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q62.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: B

Solution:

Window = N: need N+1 sequence numbers to distinguish. For N=7: need 8 (3 bits).

Subject: Computer Networks | Topic: Computer Networks | Difficulty: easy

Q64.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q65.

Answer: B

Solution:

Closures: (AB)+=ABCD, (BD)+=ABCD, (BC)+=ABCD. Minimal: AB and BD.

Subject: DBMS | Topic: DBMS | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Algorithms: 15 questions, 24 marks
Operating Systems: 8 questions, 17 marks
Engineering Mathematics: 11 questions, 16 marks
Computer Networks: 9 questions, 15 marks
Programming and Data Structures: 4 questions, 8 marks
Computer Organization: 6 questions, 6 marks
Digital Logic: 3 questions, 4 marks
Theory of Computation: 3 questions, 3 marks
DBMS: 3 questions, 3 marks
General Aptitude: 1 questions, 2 marks
Compiler Design: 2 questions, 2 marks

Difficulty Distribution:

Easy: 19 | Moderate: 36 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.